

Summary Response – Reviewer 1

Manuscript: Multi-Task Deep Learning for Food Recognition and Nutrition Estimation from Food Images

Manuscript ID: Techno-SRJ-0437

Note: Page and line references below correspond to the current 11-page rendered revision (Techno-SRJ-0437.docx). They should be rechecked once the journal template, author information, and final pagination are frozen.

No.	Reviewer Comment	Response / Action Taken	Revision Location
1	1. Reconcile the “Asian food” framing with NutritionVerse-Real, or revise the title, abstract, and framing.	We thank the reviewer for identifying this important issue. We agree that the original manuscript overstated the Asian-food characterization of NutritionVerse-Real. The revised manuscript changes the title to “Multi-Task Deep Learning for Food Recognition and Nutrition Estimation from Food Images” and reframes the study as a general food-image analysis study. Asian and Southeast Asian dietary needs remain part of the motivation, but a dedicated Cambodian/Southeast Asian dataset is now explicitly positioned as future work. The abstract, Introduction, Dataset section, Discussion, and Conclusion have been revised accordingly.	p. 1, lines 2–37; p. 2, lines 1–2; p. 3, lines 1–15; p. 10, lines 34–50
2	2. Report exact train/validation/test sizes, including image and dish counts before and after augmentation.	We added an explicit dataset-split and augmentation table. The cleaned dataset contains 225 dishes and 889 original images. The dish-level split is 180/22/23 dishes for train/validation/test, corresponding to 711/88/90 original images. Targeted augmentation added 62 training images, increasing the training set to 773 images (8.72%); validation and test remain unchanged at 88 and 90 images. The revised text also clarifies that augmentation was applied only after the dish-level split to avoid leakage.	p. 4, lines 58–69
3	3. Add efficiency evidence (latency, model size, parameter count), or soften “lightweight/real-time/mobile-deployable” claims.	We added the parameter counts reported in the final thesis: approximately 3.5 million parameters for MobileNetV2 and 25.6 million for ResNet50. We also explicitly state that this study did not measure end-to-end inference latency, model-file size, or on-device runtime. Therefore, the revised manuscript uses the evidence-supported wording “computationally efficient” and treats real-time/mobile deployment as a prospective application rather than an experimentally demonstrated result.	p. 5, lines 68–78; p. 9, lines 70–78; p. 10, lines 8–14
4	4. Benchmark F1 and R^2 against at least one prior published study.	We added a contextual comparison table and discussion. Ma et al. [16] is used for nutrient estimation because the study reports a best nutrient R^2 of 0.73 for protein on ChinaFood-100. For classification context, Yasin et al. [17] reports an F1-score of 0.87 for instance-level food classification on FoodBD v2. We explicitly explain that these are not direct head-to-head comparisons because the datasets, task definitions, label structures, and metrics differ. The comparison is therefore presented as	p. 7, lines 1–16; p. 9, lines 24–37; p. 11, lines 1–11

		contextual benchmarking rather than a claim of superiority.	
5	5. Replace numeric class indices on Fig. 5 with food-class names.	Fig. 5 has been regenerated so that the x-axis displays the actual food-class names rather than numeric class indices. The figure caption has also been revised to state that food names are shown on the x-axis.	p. 4, lines 65–68
6	6. Note the small test set (~25 dishes); consider cross-validation or significance testing.	We now report the exact independent test set size as 23 dishes and 90 images. We also explicitly state that the observed differences between MobileNetV2 and ResNet50 should be interpreted descriptively rather than as statistically conclusive evidence of superiority. Repeated cross-validation and statistical significance testing were not added because the final study protocol used a single independent dish-level split; instead, these are now stated as methodological priorities for future validation.	p. 6, lines 36–44; p. 7, lines 12–16; p. 9, lines 25–26; p. 10, lines 34–50
7	7. Fix the incomplete comparative sentence in Section 3.1.	The comparative discussion in Section 3.1 has been rewritten for grammatical completeness and scientific precision. The revised wording distinguishes observed performance differences from statistically established superiority and explains the likely influence of class imbalance and visual ambiguity.	p. 7, lines 25–32
8	8. Expand the literature review and connect prior results to this study.	The Introduction now includes additional discussion of multi-task dietary assessment and nutrient estimation. He et al. [11] is used to establish prior multi-task food/portion assessment, while Ma et al. [16] provides a quantitative nutrient-estimation benchmark. Yasin et al. [17] is also included as a recent example of fine-grained food-image analysis and nutrition estimation. The Discussion explicitly relates the present results to these studies and emphasizes differences in datasets and task definitions.	p. 2, lines 1–2; p. 7, lines 1–16; p. 9, lines 24–37; p. 11, lines 1–11

Overall Response

We sincerely thank Reviewer 1 for the constructive comments. The revision addresses the principal concern regarding the mismatch between the original Asian-food framing and the actual NutritionVerse-Real dataset, while also strengthening dataset reporting, efficiency reporting, comparative benchmarking, figure readability, uncertainty discussion, and literature positioning. The revised manuscript is intended to present the technical contribution more accurately and conservatively while retaining the motivation for future Cambodian and Southeast Asian dietary datasets.